Portfolio

Lab Experiments

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DECLARATION

Roll number F20-BS-PSY-R-1001, student of BS Psychology, session 2020 -2024 hereby declare that the matter presented in the Portfolio is my original work.

Dated: Signature:

_____ _____

PORTFOLIO COMPLETION CERTIFICATE

It is certified that the work contained in this Portfolio has been completed by F20-BS-PSY-R-1001, a student of BS Psychology, session 2020-2024, under my supervision.

Dated: Signature:

_____ _____

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Operant Conditioning & Children's Behavior

Abstract:

This study investigated the effectiveness of positive reinforcement (fixed-ratio schedule of reinforcement) in developing and increasing a 9-year-old girl's sharing of toys with her brother. Through a multi-phase design, the study observed the impact of positive reinforcement, its withdrawal/extinction, and spontaneous recovery. Results demonstrated a significant increase in sharing during the intervention phase, highlighting the effectiveness of positive reinforcement. However, when reinforcement was withdrawn, sharing decreased, suggesting the need for further intervention to maintain long-term behavior change. The follow-up phase revealed spontaneous recovery, indicating a lasting impact of the intervention. Additionally, informal assessments showed a shift in the child's attitude towards sharing, with increased willingness and enjoyment during shared play. This study provides compelling evidence for the effectiveness of positive reinforcement in modifying sharing behaviors in children. The findings contribute to understanding operant conditioning and its application in promoting prosocial behaviors and fostering healthy social interactions.

Introduction:

This study delves into the effectiveness of operant conditioning, developed by B.F. Skinner, a learning theory rooted in the principle that behavior is influenced by its consequences. proposes that behavior is shaped by its consequences. ***Operant conditioning*** *uses rewards and punishments to shape behavior by linking actions with their consequences, either strengthens or weakening those actions.*

Reinforcement *refers to anything that increases the likelihood that a response will occur.* Through operant conditioning, an association is made between a behavior and a

consequence (whether negative or positive) for that behavior. When a desired behavior is followed by a positive outcome, the likelihood of that behavior recurring increases. Conversely, when a behavior results in an undesirable consequence, it becomes less likely to be repeated.

One key component of this theory is positive reinforcement, which strengthens desired behaviors by rewarding them. ***Positive reinforcement*** *refers to the introduction of a desirable or pleasant stimulus after a behavior.* The desirable stimulus reinforces the behavior, making it more likely that the behavior will reoccur.

By understanding how positive reinforcement can influence Anaya's behavior, we aim to modify her reluctance to share the toys and foster the development of essential social skills by using fixed-ratio schedules. ***Fixed-ratio schedules*** *are those in which a response is reinforced only after a specified number of responses.* This schedule produces a high, steady rate of responding with only a brief pause after the delivery of the reinforcer.

Once a reinforcement is stopped, the response starts to decline. This is referred to as extinction, or the length of time the response continues when the reinforcement is no longer applied. In psychology, ***extinction*** *refers to the gradual weakening of a conditioned response that results in the behavior decreasing or disappearing.* In other words, the conditioned behavior eventually stops. Typically, fixed schedules of reinforcement are more prone to extinction while variable schedules are more resistant.

Spontaneous recovery *is defined as the reappearance of the conditioned response after a rest period or period of lessened response.* A behavior is displayed that was thought to be extinct.

Biodata

Name	:	Anaya
Age	:	9 years
Education	:	Grade 4
Gender	:	Female
City	:	Renala
Siblings	:	1 brother
Birth order	:	2nd

Anaya, a 9-year-old girl, hesitates to share toys with her older brother, despite willingly sharing other items. Although she has a good relationship with her brother and generally spends quality time with him. She only feels reluctant about sharing toys. Their time together is mostly pleasant, with few arguments. This study aims to explore how implementing positive reinforcement could potentially impact and adjust Anaya's behavior regarding sharing toys in question.

Within this framework, positive reinforcement (fixed-ratio schedule of reinforcement) plays a crucial role in strengthening desirable behaviors. Positive reinforcement can take various forms, including verbal praise, and tangible rewards (chocolates). Providing positive reinforcement after every third occurrence of the desired behavior, using a fixed-ratio schedule, strengthens the connection between the behavior and the reward. This increased association makes it more probable that the desired behavior will be repeated in the future.

In the context of Anaya's reluctance to share, positive reinforcement can be implemented by offering her rewards such as verbal praise or chocolates every third time she allows her brother to play with the toys. Over time, this positive association between sharing and the reward will increase the likelihood of Anaya's sharing the toys in the future, fostering positive social interactions with her brother and contributing to her overall development.

Objectives

1. To observe and document Anaya's natural sharing behavior with her brother during a baseline phase.
2. To monitor the frequency of sharing behavior through a checklist and real-time observation throughout the experiment.
3. To assess the effectiveness of positive reinforcement in increasing Anaya's sharing behavior.
4. To observe if the extinction occurs when the positive reinforcement is withdrawn.
5. To determine if spontaneous recovery of sharing behavior occurs after the extinction.

Methodology

This study employed a single-subject research design, a powerful method for investigating individual behavior. Unlike traditional research designs involving multiple subjects, this approach focuses on a single individual - Anaya. This approach enables us to conduct a granular and in-depth examination of individual trajectories in response to interventions.

This study further employed a multi-phase design, encompassing a baseline phase, an intervention phase, an extinction phase, and a follow-up phase. Each phase lasted one week, providing a structured framework for observing and analyzing the changes in Anaya's sharing behavior.

1. During the **baseline phase**, no intervention was implemented, allowing us to establish a baseline level of Anaya's sharing behavior toward her brother. This served as a

reference point for evaluating the effectiveness of the intervention in the subsequent phases.

2. In the **intervention** phase, positive reinforcement was provided after each third instance of Anaya's sharing the toys with her brother. This reinforcement consisted of verbal praise or chocolates, thereby increasing the likelihood of Anaya's repeating the desired behavior (sharing the toys).
3. The **extinction** phase involved withdrawing the positive reinforcement. This phase aimed to observe whether Anaya's sharing behavior would decrease back to baseline levels or remain elevated due to the learned association established during the intervention phase.
4. Finally, the **follow-up** phase helped us determine if spontaneous recovery occurred, meaning whether Anaya would resume sharing the toys with receiving rewards after a week-long gap of the extinction phase.

Assessment

Evaluating the effectiveness of interventions requires a comprehensive assessment strategy encompassing both formal and informal methods. This case study explored the impact of positive reinforcement (fixed-ratio schedule of reinforcement) on Anaya's sharing behavior using a combination of formal and informal assessments, providing a holistic picture of behavioral change.

Formal Assessment

In context of this study, formal assessment focused on the behavior of interest - Anaya's sharing of the toys with her brother. A structured checklist was used throughout the study to track the situation and instances of sharing the toys with her brother. The checklist included the following columns:

- **Phase:** Stages of the experiment designed to assess the effectiveness of the reinforcement.
- **Situation:** Description of the situation (e.g., free play, sibling interaction).
- **Sharing behavior:** Whether Anaya shared toys or not (yes/no).
- **Free Play:** Anaya is playing with the toy independently.
- **Sibling Interaction:** Anaya is interacting with her brother while playing with the toy.
- **Positive reinforcement:** Type of positive reinforcement provided (e.g., verbal praise, chocolate)

Sharing Behavior Checklist

Phase	*Week*	*Situation*	*Sharing Behavior*	*Sibling Interaction*	*Positive Reinforcement*
Baseline	***1***	Sibling Interaction	No	5 min (Avg./Day)	N/A
Intervention	***2***	Sibling Interaction	Yes	28 min (Avg./Day)	Verbal Praise, Chocolate
Extinction	***3***	Sibling Interaction	≈ No[1]	12 min (Avg./Day)	N/A
Follow-up	***4***	Sibling Interaction	Yes	25 min (Avg./Day)	Verbal Praise, Chocolate

Baseline - Week 1

Day	*Situation*	*Sharing Behavior*	*Sibling Interaction*	*Positive Reinforcement*
Monday	Free Play	No	0 min (Avg./Day)	N/A
Tuesday	Sibling Interaction	No	15 min (Avg./Day)	N/A
Wednesday	Free Play	No	0 min (Avg./Day)	N/A
Thursday	Free Play	No	0 min (Avg./Day)	N/A
Friday	Free Play	No	0 min (Avg./Day)	N/A
Saturday	Free Play	No	0 min (Avg./Day)	N/A
Sunday	Sibling Interaction	No	20 min (Avg./Day)	N/A

[1] *Extinction phase initially saw continued sharing during sibling interaction for two days, likely due to residual intervention effects. However, without further reinforcement, sharing behavior extinguished.*

Intervention - Week 2

Day	*Situation*	*Sharing Behavior*	*Sibling Interaction*	*Positive Reinforcement*
Monday	Free Play	No	0 min (Avg./Day)	N/A
Tuesday	Sibling Interaction	Yes	45min (Avg./Day)	Verbal Praise, Chocolate
Wednesday	Free Play	No	0 min (Avg./Day)	N/A
Thursday	Sibling Interaction	Yes	40 min (Avg./Day)	Verbal Praise, Chocolate
Friday	Sibling Interaction	Yes	30 min (Avg./Day)	Verbal Praise, Chocolate
Saturday	Sibling Interaction	Yes	35 min (Avg./Day)	Verbal Praise, Chocolate
Sunday	Sibling Interaction	No	50 min (Avg./Day)	Verbal Praise, Chocolate

Extinction - Week 3

Day	*Situation*	*Sharing Behavior*	*Sibling Interaction*	*Positive Reinforcement*
Monday	Sibling Interaction	Yes	40 min (Avg./Day)	N/A
Tuesday	Sibling Interaction	Yes	25 min (Avg./Day)	N/A
Wednesday	Sibling Interaction	No	15 min (Avg./Day)	N/A
Thursday	Sibling Interaction	No	5 min (Avg./Day)	N/A
Friday	Free Play	No	0 min (Avg./Day)	N/A
Saturday	Free Play	No	0 min (Avg./Day)	N/A
Sunday	Free Play	No	0 min (Avg./Day)	N/A

Follow up - Week 4

Day	*Situation*	*Sharing Behavior*	*Sibling Interaction*	*Positive Reinforcement*
Monday	Free Play	No	0 min (Avg./Day)	N/A
Tuesday	Sibling Interaction	Yes	35 min (Avg./Day)	Verbal Praise, Chocolate
Wednesday	Free Play	No	0 min (Avg./Day)	N/A
Thursday	Sibling Interaction	Yes	30 min (Avg./Day)	Verbal Praise, Chocolate
Friday	Sibling Interaction	Yes	25 min (Avg./Day)	Verbal Praise, Chocolate
Saturday	Sibling Interaction	Yes	40 min (Avg./Day)	Verbal Praise, Chocolate
Sunday	Sibling Interaction	Yes	45 min (Avg./Day)	Verbal Praise, Chocolate

Informal Assessment

Informal assessments provide qualitative data through observations. During all phases of the study, the researcher directly observed Anaya's interactions with her brother to assess her sharing behavior in real-time. This involved observing her nonverbal cues, communication, and willingness to share the toys. The researcher used a pen and paper to take notes during these observations, focusing on any changes in Anaya's behavior in response to the intervention and investigating the impact of positive reinforcement on Anaya's sharing behavior.

1. **Baseline (Week 1):** No intervention was implemented. Anaya was good at communication but reluctant to share toys.
2. **Intervention (Week 2):** Anaya showed a willingness to share toys with her brother, doing calmly and openly. Positive reinforcement was given right after every third occasion she shared, encouraging this behavior.
3. **Extinction (Week 3):** When positive reinforcement vanished, a subtle shift appeared in Anaya's interactions. While she still interacted with her brother, the eagerness and joy that had characterized her sharing during the intervention phase seemed to soften, replaced by a hint of hesitation.
4. **Follow-up (Week 4):** Anaya demonstrated a spontaneous return to sharing toys with her brother when reinforcement was reintroduced. Anaya's sharing not only resumed instantly but also maintained a relaxed and welcoming demeanor, indicating a deeper change in their relationship. Spontaneously offering toys and exchanging smiles, their playtime exuded a newfound warmth and receptivity.

Pre-assessment

Real-time observations were conducted during the **baseline phase** (Week 1) to observe Anaya's natural interactions with her brother regarding the sharing of toys. Prior to the intervention, Anaya exhibited reluctance to share the toys with her brother. Observations revealed hesitance and occasional frustration when her brother attempted to play with it. The researcher also observed the potential fear of losing the toys or difficulty in sharing due to its perceived specialness.

Post-assessment

Real-time observations were continued during the **intervention phase** (Week 2) to monitor any changes in Anaya's behavior and interactions after receiving positive reinforcement. Following the intervention, Anaya displayed an increased willingness to share the toys. Observations noted more positive interactions with her brother, including negotiation and compromise during playtime, improved communication, and a sense of enjoyment derived from sharing experiences with her sibling.

Conclusion

This single-subject study investigated the effectiveness of positive reinforcement in modifying a 9-year-old girl's sharing behavior towards her brother. Utilizing a structured research design, the study tracked the impact of positive reinforcement (fixed-ratio schedule of reinforcement) on the frequency of the desired behavior.

The findings revealed a significant increase in sharing behavior after implementing positive reinforcement. Anaya shared toys with her brother during the intervention phase as compared to the baseline phase(during which she always felt hesitant in sharing toys). However, upon withdrawal of reinforcement, sharing decreased back towards baseline levels,

highlighting the need for further interventions to sustain the positive behavior change. Interestingly, the follow-up period demonstrated spontaneous recovery, with Anaya showcasing continued sharing behavior.

Informal assessments further endorsed the findings. Observations before the intervention revealed Anaya's reluctance to share, while during the intervention phase, observation indicated increased willingness and improved communication during shared playtime. This suggests a positive shift in Anaya's attitude towards sharing, potentially influencing her social interactions and relationships.

Overall, this study provides valuable evidence for the effectiveness of positive reinforcement in promoting sharing behavior in children. The significant improvement in Anaya's sharing demonstrates the potential of this intervention for fostering positive social skills.

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